

Statistical Analysis

Audience Reviews of 3 Idiots

Sathi Khanra

Department of Computer Science
AI & Machine Learning

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Movie Introduction

About the Film

3 Idiots (2009) is one of the most influential Bollywood films, directed by Rajkumar Hirani. The movie explores themes of education, innovation, friendship, and social expectations while delivering a powerful message about learning and personal growth.

Analytical Relevance

- Strong public recognition and popularity.
- Large volume of audience reviews available.
- Diverse viewer demographics.
- Ideal case study for sentiment and statistical analysis.

Project Overview & Research Motivation

Research Objective

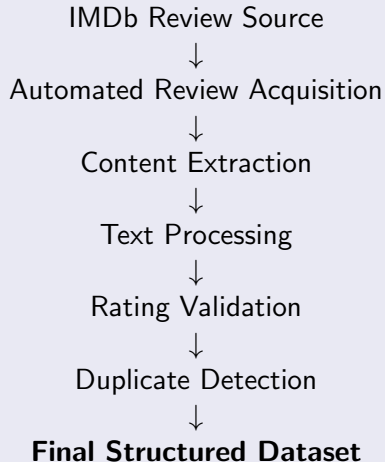
This project investigates audience perceptions of the Bollywood film **3 Idiots** through quantitative analysis of IMDb review ratings. By applying descriptive statistical techniques and sentiment classification, the study aims to uncover patterns in audience satisfaction, rating behavior, and collective opinion formation.

Why This Study Matters

- Audience reviews represent real viewer experiences.
- Statistical analysis converts opinions into measurable insights.
- Understanding audience behavior supports entertainment analytics.
- Data-driven findings provide objective evaluation of movie reception.

Dataset Creation Pipeline

Review Collection Workflow



Pipeline Highlights

- Reviews collected from IMDb audience pages.
- Ratings and review texts extracted separately.
- Duplicate entries removed.
- Invalid ratings filtered.
- Structured dataset prepared for statistical analysis.
- Data validated before visualization and interpretation.

Cleaning Results

- Raw Dataset Size: **2228 Reviews**
- Final Clean Dataset Size: **1043 Reviews**
- Duplicate Reviews Removed: **1141**
- Invalid / Missing Ratings Removed: **92**
- Rating Scale Successfully Validated (1–10)
- Dataset Prepared for Statistical Analysis

Outcome

A high-quality dataset containing **1043 validated audience reviews** was retained for descriptive statistical analysis, sentiment evaluation, and visualization.

Key Analytical Questions

- ① How satisfied are audiences with 3 Idiots?
- ② What rating values occur most frequently?
- ③ How consistent are audience opinions?
- ④ Is audience perception predominantly positive?
- ⑤ Does the rating distribution indicate consensus?
- ⑥ What behavioral patterns emerge from review data?

Sentiment Analysis

Purpose of Sentiment Analysis

Sentiment analysis is a technique used to classify audience opinions into meaningful categories based on their expressed satisfaction level.

For this study, audience ratings were transformed into sentiment labels to better understand the emotional response toward the movie.

Sentiment Classification Rules

Rating Range	Sentiment
8 – 10	Positive
5 – 7	Neutral
1 – 4	Negative

Statistical Overview

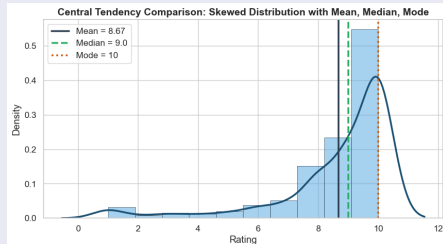
Central Tendency

- Mean: **8.67**
- Median: **9.00**
- Mode: **10**

Dispersion Measures

- Range: **9**
- Variance: **4.28**
- Std. Deviation: **2.07**

Distribution Visualization



Distribution Characteristics

- Skewness: **-2.20**
- Kurtosis: **4.72**
- Sample Size: **1043**

Range Analysis

Definition

The range measures the overall spread of audience ratings by calculating the difference between the highest and lowest rating.

$$\text{Range} = \text{Maximum Rating} - \text{Minimum Rating}$$

Calculation

$$\text{Range} = 10 - 1 = 9$$

Interpretation

- The rating range of **9** indicates a wide spread of audience opinions.
- Both extremely positive and extremely negative reviews exist.
- Although most viewers rated the movie highly, a small group expressed lower satisfaction.

Visualization Overview

Why Visualization Matters

- Converts statistics into visual insights.
- Simplifies complex patterns.
- Reveals trends and variability.
- Supports statistical evidence.
- Improves communication.

Role in Audience Analytics

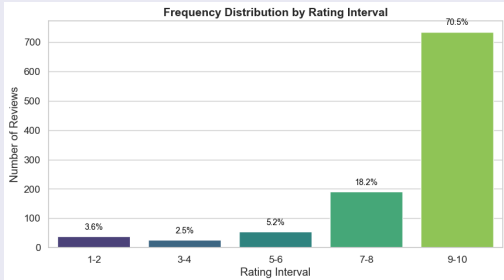
- Reveals sentiment patterns.
- Detects audience consensus.
- Highlights rating concentration.
- Identifies skewness.
- Explains audience behavior.

Key Insight

Visual analysis complements descriptive statistics by revealing patterns that are difficult to observe through numerical summaries alone.

Frequency Distribution by Rating Interval

Rating Interval Distribution



Interpretation

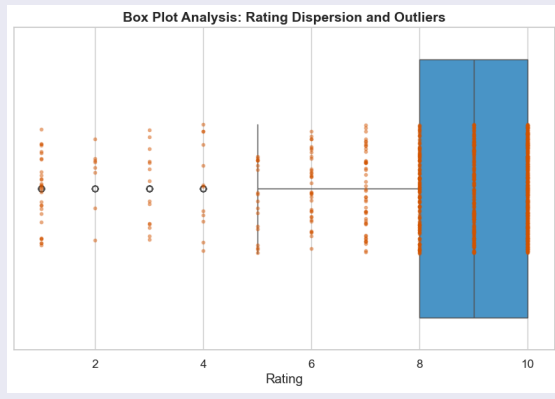
- Ratings are grouped into five intervals.
- The **9–10** interval has the highest frequency.
- The **7–8** interval shows additional positive sentiment.
- Lower rating intervals contain relatively few reviews.
- Overall, audience opinion is strongly favorable.

Key Insight

Most reviews fall within the **9–10** interval, indicating overwhelming audience approval and strong positive reception.

Box Plot Analysis

Rating Distribution



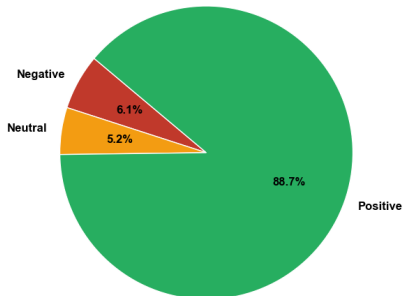
Interpretation

- Ratings are concentrated in the higher score range.
- The median lies close to the upper quartile, indicating strong audience satisfaction.
- A few low-rating outliers represent limited negative opinions.
- The narrow interquartile spread suggests consistent audience perception.
- The distribution supports widespread positive approval of the movie.

Sentiment Composition

Sentiment Distribution

Sentiment Composition: Negative, Neutral, and Positive Reviews



Interpretation

- Positive sentiment dominates the review landscape.
- Neutral reviews constitute a relatively small proportion of audience responses.
- Negative sentiment represents only a limited fraction of viewers.
- The sentiment composition indicates strong audience appreciation and favorable emotional engagement.
- The distribution provides evidence of widespread acceptance and popularity of the film.

Statistical Insight Summary

Statistic	Value	Interpretation
Mean	8.67	High overall audience satisfaction and approval.
Median	9	Typical audience opinion is strongly positive.
Mode	10	Most viewers awarded the highest possible rating.
Range	9	Presence of both highly positive and limited negative opinions.
Variance	4.28	Moderate spread around the average rating.
Std. Deviation	2.07	Ratings remain relatively concentrated near the mean.
Skewness	-2.20	Strong left-skewness indicates dominance of high ratings.
Kurtosis	4.72	Ratings are highly concentrated with a pronounced peak.

Audience Behavior Analytics

Key Behavioral Insights

- Mean (8.67) and Median (9) indicate high audience satisfaction.
- Mode (10) shows that perfect ratings are most common.
- Negative skewness (-2.20) confirms dominance of high ratings.
- Kurtosis (4.72) suggests strong audience consensus.
- The 9–10 interval contains the majority of reviews.
- Positive opinions greatly outweigh negative responses.

Audience Psychology

High ratings and strong consensus indicate that **3 Idiots** was widely appreciated and emotionally resonated with its audience.

Conclusion

Key Findings

- Audience reception of **3 Idiots** is overwhelmingly positive.
- High mean, median, and mode values indicate strong viewer satisfaction.
- Negative skewness confirms that high ratings dominate the distribution.
- The 9–10 rating interval contains the majority of reviews, reflecting widespread approval.
- Statistical and visual analyses consistently support a strong audience consensus.

Final Remark

The findings demonstrate that **3 Idiots** resonated strongly with its audience, achieving high levels of satisfaction, appreciation, and positive engagement. The statistical evidence confirms the film's enduring popularity and favorable reception.

Thank You

Questions, Suggestions
or Discussion

Statistical Analysis of Audience Reviews of 3 Idiots

Sathi Khanra